

Tyler Stoen

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Statistician and Data Scientist with a strong foundation in statistical methods and machine learning. Motivated by finding the best solution to a problem by building, comparing, and refining models to optimize performance. Proven experience handling high-dimensional data for model development and evaluation, specifically in biological settings.

EDUCATION

California Polytechnic State University, San Luis Obispo

MS in Statistics | GPA: 4.0

San Luis Obispo, CA

Sept 2025 – June 2026

- Thesis: "[Discovering Differentially Correlated Sets Across Multiple Treatments](#)"

California Polytechnic State University, San Luis Obispo

BS in Statistics, Cross-Disciplinary Minor in Data Science | Summa cum laude

San Luis Obispo, CA

Sept 2021 – June 2025

Relevant Coursework: Statistical Learning, Generalized Linear Models, Bayesian Methods, Design of Experiments, Time Series, Adv. Statistical Computing, Distributed Computing, Graduate Consulting, Stochastic Processes, Data Science Process & Ethics,

WORK EXPERIENCE

California Polytechnic State University, San Luis Obispo – Statistics

San Luis Obispo, CA

Student Researcher

June 2025 – September 2025

- Explored and compared candidate affinity measures for detecting differential correlation across three experimental groups.
- Formulated a variance-based extension of pairwise differential correlation metrics for multi-group settings.
- Implemented procedure from existing statistical literature on multi-group correlation and p-value adjustment methods.

Stanford University School of Medicine – Davis Lab

Stanford, CA

Student Intern – Data Science

June 2023 – September 2023

- Cleaned, preprocessed, and visualized mass cytometry data using R.
- Implemented and helped improve a user-contributed R package built for analyzing cytometry data.
- Trained and tested several different distance-based classification algorithms on mass cytometry data.
- Determined a new distance method that classified patient cell types up to 5 percentage points more accurately than previously used methods.

PROJECTS

Microorganism Diversity in Vineyard Soil

Jan 2025 – June 2025

California Polytechnic State University, San Luis Obispo

San Luis Obispo, CA

- Partnered with a client to analyze vineyard soil data, designing a tailored analytical approach around their research questions.
- Modeled relationships between soil characteristics and microbial communities across different soil treatments.
- Identified the key drivers of microbial diversity and compared microorganism functionality between treatments.
- Delivered actionable recommendations to guide the client's vineyard soil management decisions.

Plastic Waste

April 2026 – June 2026

California Polytechnic State University, San Luis Obispo

San Luis Obispo, CA

- Authored a formal statistical report investigating regional, corporate, and plastic-type patterns in global waste data, integrating API-sourced GDP, population, and geographic covariates.
- Built an interactive Quarto dashboard enabling world and country-level exploration of recorded plastic waste metrics.
- Developed a modular R package to streamline all data preprocessing and exploration.

Email Spam Detection

Nov 2025– Dec 2025

California Polytechnic State University, San Luis Obispo

San Luis Obispo, CA

- Built a natural language processing pipeline to classify text and email messages as spam or ham using TF-IDF features.
- Trained and tuned logistic regression, LASSO, decision tree, and random forest models, comparing performance with 10-fold cross-validation.
- Identified the most predictive words for spam classification through model interpretation.

Image Classification

Nov 2024 – Dec 2024

California Polytechnic State University, San Luis Obispo

San Luis Obispo, CA

- Built an EfficientNet-B0 model as a baseline image classifier to distinguish photos taken by two photographers, achieving 92% accuracy.
- Engineered interpretable image features related to contents and composition and trained a separate explanatory neural network on them.
- Combined the two models into an ensemble, improving test accuracy to 96% over the deep-learning baseline.

SKILLS

Programming: Python (TensorFlow, PyTorch), R, SAS, Java, JMP, Hadoop, Scala/Spark, SQL

Technical: LaTeX (Overleaf/R Markdown), Quarto, Tableau, Microsoft Excel, Git

INTERESTS

Music: Completed the Advanced Level of the Certificate of Merit test for piano with state honors. Continue to play for enjoyment.

Swimming: Lifetime swimmer and Cal Poly Swim Club Alumni. Previously served on the club board as Meet Coordinator (2024–2025).